ETH zürich

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Algorithms and Probability **Theory Sheet T2** **FS 2026**

The solutions for this sheet must be submitted on Moodle by 19 March, 15:59.

The following exercise is worth 2 bonus points.

Exercise T2.1 – *Choosing Seminars*

Every student in your semester applied for exactly three different seminars. Two of the seminars received exactly 40 applications. All other seminars received less than 40 applications. Is it possible to assign each student to one of the three seminars they applied to such that there are at most 13 students in every seminar? How about at most 12 students per seminar?

The following exercise does not give bonus points.

Exercise T2.2 – *Augmenting path of length* k

Let $G = (V, E)$ be a graph. Let M and M' be two matchings in G.

(a) Show that there are at least $|M'| - |M|$ vertex-disjoint M-augmenting paths in G.

(b) Show that if every M-augmenting path has length at least $2k + 1$ then $|M'| \leq \frac{k+1}{k}|M|$.

(c) Is the bound in (b) tight (i.e., can it happen that $|M'| = \frac{k+1}{k}|M|$)?